



Guia 3– VLANs

Objectivos:

- Definição de VLANs
- Routing entre VLANs
- Utilização de switches L2 e L3
- Desenho e interligação de redes de Acesso e Distribuição
- Estabelecimento e configuração de trunks

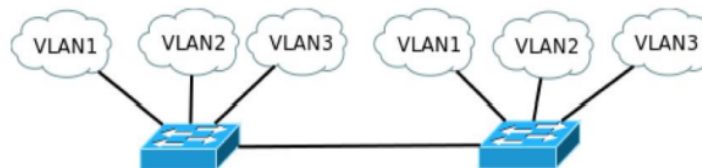
1 - Definição de VLANs

1. **(LEIA O EXERCÍCIO TODO PRIMEIRO ANTES DE COMEÇAR A RESOLVÊ-LO!!!)**

Usando o GNS3, monte a rede da figura. Em vez das núvens pode utilizar VPCs ou outra entidade qualquer que simule um host (aconselha-se, por exemplo, um router c3725 configurado com um endereço IP na interface).

Configure três VLANs nos switches:

- Ports 1-2: VLAN1 (sub-rede 10.1.1.0/24)
- Ports 3-4: VLAN2 (sub-rede 10.2.2.0/24)
- Ports 5-6: VLAN3 (sub-rede 10.3.3.0/24)
- Ports 7-8: Inter-switch/Tagged/802.1Q (native VLAN 1)



NOTA:

Para replicar um switch Layer2 no GNS3 aconselha-se antes a utilização de um router c3725 configurado com um modulo de switching (NM- 16SW). Esse modulo adiciona interfaces de switching ao router, que têm o protocolo IP desligado (portos f1/0 a f1/15). Esta combinação é conhecida por “EtherSwitch router”.

Nota: O switch básico do GNS3 “Ethernet Switch” não suporta o Spanning Tree Protocol.

Configurar um “Ethernet Switch” do GNS3: Usar a interface gráfica.

Configurar um “EtherSwitch router” como sendo um switch L2:

```
EtherSwitch# vlan database
EtherSwitch(vlan)# vlan 1
EtherSwitch(vlan)# vlan 2
EtherSwitch(vlan)# vlan 3
EtherSwitch(vlan)# exit
EtherSwitch# configure terminal
EtherSwitch(config)# no ip routing
EtherSwitch(config)# interface f1/1
EtherSwitch(config-if)# switchport mode access
```



```
EtherSwitch(config-if)# switchport access vlan 1
EtherSwitch(config-if)# interface f1/2
EtherSwitch(config-if)# switchport mode access
EtherSwitch(config-if)# switchport access vlan 1
EtherSwitch(config-if)# interface range fastEthernet 1/3 - 4
EtherSwitch(config-if-range)# switchport mode access
EtherSwitch(config-if-range)# switchport access vlan 2
EtherSwitch(config-if-range)# interface range fastEthernet 1/5 - 6
EtherSwitch(config-if-range)# switchport mode access
EtherSwitch(config-if-range)# switchport access vlan 3
EtherSwitch(config-if-range)# interface range fastEthernet 1/7 - 8
EtherSwitch(config-if-range)# switchport mode trunk          !Defines as Trunk
port
EtherSwitch(config-if-range)# switchport trunk encapsulation dot1q !By
default all
```

Note: Comando a usar para mostrar as VLANs existentes: show vlan-switch

Soluções para problemas:

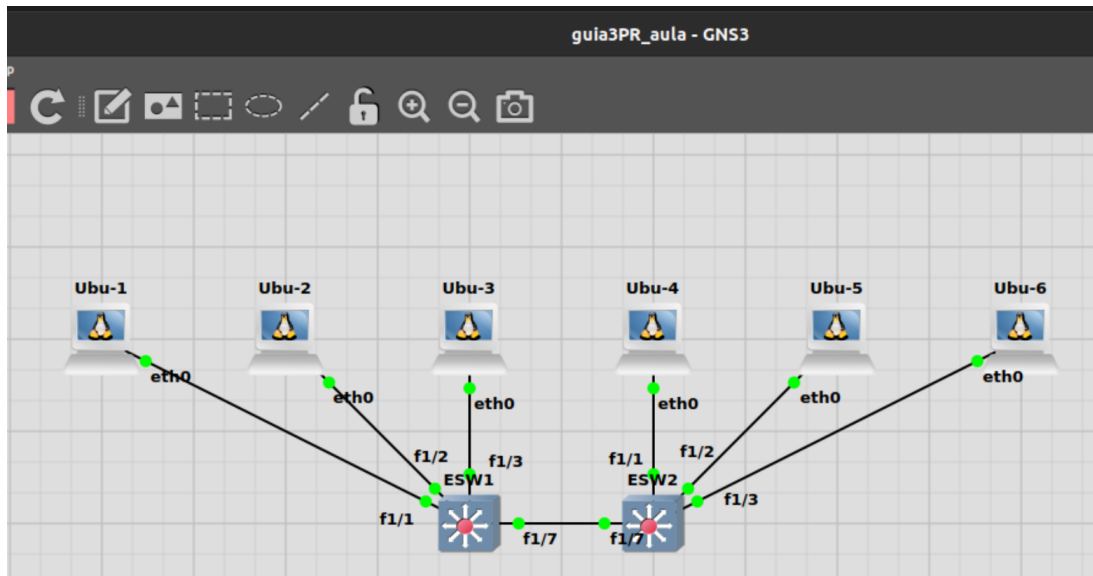
Nota: Se aparecer “(no device available)” tente adicionar pelo menos mais 1MB ao PCMCIA slot 0, na configuração do router (configuração gráfica do router no GNS3).

Troubleshooting 2: Verifique se todas as interfaces com ligações estão ativas, com o comando:
show ip interface brief
Coloque terminais (VPCs, hosts, núvens, etc.) nas diferentes VLANs e teste a conectividade.

2. Grave as configurações dos equipamentos, bem como o projecto.

Definição de VLANs

1.



Ubu-1 e Ubu-4 pertencem à VLAN 1.

Ubu-2 e Ubu-5 pertencem à VLAN 2.

Ubu-3 e Ubu-6 pertencem à VLAN 3.

Configuração de cada host:

Vlan1

Ubu-1:

```
root@Ubu-1:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#
# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.1.1.1
    netmask 255.255.255.0
    gateway 192.168.0.1
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname Ubu-1
```

Ubu-4:

```
root@Ubu-4:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#
# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.1.1.2
    netmask 255.255.255.0
    gateway 192.168.0.1
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname Ubu-4
```

Vlan 2

Ubu-2:

```
root@Ubu-2:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#
# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.2.2.1
    netmask 255.255.255.0
    gateway 192.168.0.1
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname Ubu-2
```

Ubu-5:

```
root@Ubu-5:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#
# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.2.2.2
    netmask 255.255.255.0
    gateway 192.168.0.1
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname Ubu-5
```

Vlan 3

Ubu-3:

```
root@Ubu-3:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#
# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.3.3.1
    netmask 255.255.255.0
#    gateway 192.168.0.1
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname Ubu-3
```

Ubu-6:

```
root@Ubu-6:~# cat /etc/network/interfaces
#
# This is a sample network config, please uncomment lines to configure the network
#
# Uncomment this line to load custom interface files
# source /etc/network/interfaces.d/*

# Static config for eth0
auto eth0
iface eth0 inet static
    address 10.3.3.2
    netmask 255.255.255.0
#    gateway 192.168.0.1
#    up echo nameserver 192.168.0.1 > /etc/resolv.conf

# DHCP config for eth0
#auto eth0
#iface eth0 inet dhcp
#    hostname Ubu-6
```

```
ESW1#show vlan-switch
```

VLAN	Name	Status	Ports
1	default	active	Fa1/0, Fa1/1, Fa1/4, Fa1/5 Fa1/6, Fa1/7, Fa1/8, Fa1/9 Fa1/10, Fa1/11, Fa1/12, Fa1/13 Fa1/14, Fa1/15
2	VLAN0002	active	Fa1/2
3	VLAN0003	active	Fa1/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	1002	1003
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	1	1003
1003	tr	101003	1500	1005	0	-	-	srb	1	1002
1004	fdnet	101004	1500	-	-	1	-	ibm	0	0
1005	trnet	101005	1500	-	-	1	-	ibm	0	0

```
ESW2#show vlan-switch
```

VLAN	Name	Status	Ports
1	default	active	Fa1/0, Fa1/1, Fa1/3, Fa1/4 Fa1/5, Fa1/6, Fa1/7, Fa1/8 Fa1/9, Fa1/10, Fa1/11, Fa1/12 Fa1/13, Fa1/14, Fa1/15
2	VLAN0002	active	Fa1/2
3	VLAN0003	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLAN	Type	SAID	MTU	Parent	RingNo	BridgeNo	Stp	BrdgMode	Trans1	Trans2
1	enet	100001	1500	-	-	-	-	-	1002	1003
2	enet	100002	1500	-	-	-	-	-	0	0
3	enet	100003	1500	-	-	-	-	-	0	0
1002	fddi	101002	1500	-	-	-	-	-	1	1003
1003	tr	101003	1500	1005	0	-	-	srb	1	1002
1004	fdnet	101004	1500	-	-	1	-	ibm	0	0
1005	trnet	101005	1500	-	-	1	-	ibm	0	0

Configurações de cada EtherSwitch (fazendo show run):

ESW1#sh run

Building configuration...

Current configuration : 2628 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

no service dhcp

!

hostname ESW1

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

memory-size iomem 5

no ip routing

no ip icmp rate-limit unreachable

no ip cef

!

```
!  
!  
!  
no ip domain lookup  
ip auth-proxy max-nodata-conns 3  
ip admission max-nodata-conns 3  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
no ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
no ip route-cache  
shutdown  
duplex auto
```

```
speed auto
!
interface FastEthernet1/0
duplex full
speed 100
!
interface FastEthernet1/1
duplex full
speed 100
!
interface FastEthernet1/2
switchport access vlan 2
duplex full
speed 100
!
interface FastEthernet1/3
switchport access vlan 3
duplex full
speed 100
!
interface FastEthernet1/4
duplex full
speed 100
!
interface FastEthernet1/5
duplex full
speed 100
!
interface FastEthernet1/6
duplex full
speed 100
!
interface FastEthernet1/7
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/8
duplex full
speed 100
!
interface FastEthernet1/9
duplex full
speed 100
!
```

```
interface FastEthernet1/10
duplex full
speed 100
!
interface FastEthernet1/11
duplex full
speed 100
!
interface FastEthernet1/12
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
duplex full
speed 100
!
interface FastEthernet1/15
duplex full
speed 100
!
interface Vlan1
no ip address
no ip route-cache
shutdown
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
```

!
!
!
!
!

banner exec ^C

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

^C

!

line con 0

exec-timeout 0 0

privilege level 15

logging synchronous

line aux 0

exec-timeout 0 0

privilege level 15

logging synchronous

line vty 0 4

login

!

!

end

ESW2#sh run

Building configuration...

Current configuration : 2628 bytes

!

version 12.4

service timestamps debug datetime msec

service timestamps log datetime msec

no service password-encryption

no service dhcp

!

hostname ESW2

!

boot-start-marker

boot-end-marker

!

!

no aaa new-model

memory-size iomem 5

no ip routing

no ip icmp rate-limit unreachable

no ip cef

!

!

!

!

no ip domain lookup

ip auth-proxy max-nodata-conns 3

ip admission max-nodata-conns 3

!

!

!

!

!

!

!

!

!

!

!

!

!

!

```
!  
vtp file nvram:vlan.dat  
  
!  
!  
ip tcp synwait-time 5  
!  
!  
!  
!  
!  
interface FastEthernet0/0  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
no ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet0/1  
description *** Unused for Layer2 EtherSwitch ***  
no ip address  
no ip route-cache  
shutdown  
duplex auto  
speed auto  
!  
interface FastEthernet1/0  
duplex full  
speed 100  
!  
interface FastEthernet1/1  
duplex full  
speed 100  
!  
interface FastEthernet1/2  
switchport access vlan 2  
duplex full  
speed 100  
!  
interface FastEthernet1/3  
switchport access vlan 3  
duplex full  
speed 100  
!
```

```
interface FastEthernet1/4
duplex full
speed 100
!
interface FastEthernet1/5
duplex full
speed 100
!
interface FastEthernet1/6
duplex full
speed 100
!
interface FastEthernet1/7
switchport mode trunk
duplex full
speed 100
!
interface FastEthernet1/8
duplex full
speed 100
!
interface FastEthernet1/9
duplex full
speed 100
!
interface FastEthernet1/10
duplex full
speed 100
!
interface FastEthernet1/11
duplex full
speed 100
!
interface FastEthernet1/12
duplex full
speed 100
!
interface FastEthernet1/13
duplex full
speed 100
!
interface FastEthernet1/14
duplex full
speed 100
!
```

```
interface FastEthernet1/15
duplex full
speed 100
!
interface Vlan1
no ip address
no ip route-cache
shutdown
!
ip forward-protocol nd
!
!
no ip http server
no ip http secure-server
!
no cdp log mismatch duplex
!
!
!
!
control-plane
!
!
!
!
!
!
!
!
!
banner exec ^C
```

This is a normal Router with a SW module inside (NM-16ESW)
It has been preconfigured with hard coded speed and duplex

To create vlans use the command "vlan database" from exec mode
After creating all desired vlans use "exit" to apply the config

To view existing vlans use the command "show vlan-switch brief"

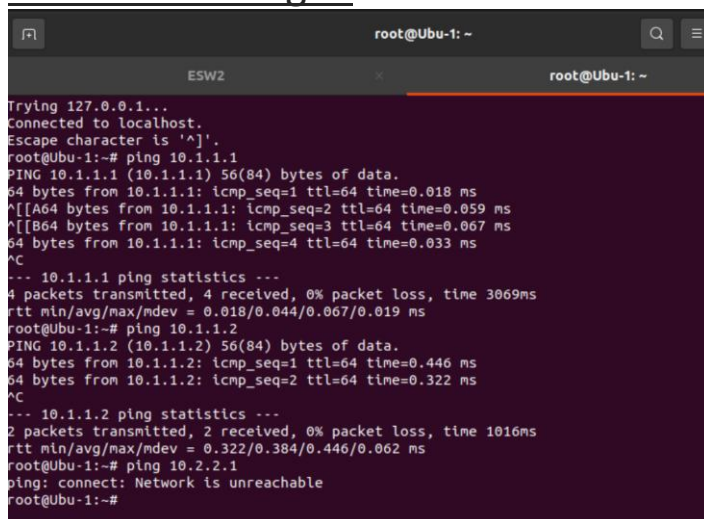
Warning: You are using an old IOS image for this router.
Please update the IOS to enable the "macro" command!

```

^C
!
line con 0
exec-timeout 0 0
privilege level 15
logging synchronous
line aux 0
exec-timeout 0 0
privilege level 15
logging synchronous
line vty 0 4
login
!
!
end

```

Troubleshooting 2:



```

root@Ubu-1: ~
ESW2
root@Ubu-1: ~
Trying 127.0.0.1...
connected to localhost.
Escape character is '^]'.
root@Ubu-1:~# ping 10.1.1.1
PING 10.1.1.1 (10.1.1.1) 56(84) bytes of data.
64 bytes from 10.1.1.1: icmp_seq=1 ttl=64 time=0.018 ms
^[[A64 bytes from 10.1.1.1: icmp_seq=2 ttl=64 time=0.059 ms
^[[B64 bytes from 10.1.1.1: icmp_seq=3 ttl=64 time=0.067 ms
64 bytes from 10.1.1.1: icmp_seq=4 ttl=64 time=0.033 ms
^C
--- 10.1.1.1 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3069ms
rtt min/avg/max/mdev = 0.018/0.044/0.067/0.019 ms
root@Ubu-1:~# ping 10.1.1.2
PING 10.1.1.2 (10.1.1.2) 56(84) bytes of data.
64 bytes from 10.1.1.2: icmp_seq=1 ttl=64 time=0.446 ms
64 bytes from 10.1.1.2: icmp_seq=2 ttl=64 time=0.322 ms
^C
--- 10.1.1.2 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1016ms
rtt min/avg/max/mdev = 0.322/0.384/0.446/0.062 ms
root@Ubu-1:~# ping 10.2.2.1
ping: connect: Network is unreachable
root@Ubu-1:~#

```

Vlan 1 comunica entre si através do trunk entre ESW1 e ESW2.
Vlans diferentes não têm conectividade.

2. Guardado.